

Day 3: Switch and Router

Configuration BasicsTheory

Task 1 :- Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

ANS :-

To create the Sales and Support VLANs on a switch in Cisco Packet Tracer, follow these steps using either the Command Line Interface (CLI) or the Graphical User Interface (GUI).

1.Add the Switch to the Canvas : Workspace Setup.

1. Open Cisco Packet Tracer.
2. Click on Network Devices (bottom-left) → Switches.
3. Drag a 2960 switch onto the workspace center.

2.Configure VLANs via CLI : Recommended / Command Line Method.

1. Click on the switch to open its configuration window and switch to the **CLI** tab.
2. Press Enter to get the Switch> prompt.
3. Run the following commands to enter configuration mode and create both VLANs:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# end
```

3.Alternative: Configure VLANs via GUI : No Commands Required.

1. Click on the switch and select the Config tab.
2. In the left panel under SWITCHING, click on VLAN Database.
3. Enter the details for the first VLAN:
 - VLAN Number: 10
 - VLAN Name: Sales
 - Click Add.
4. Enter the details for the second VLAN:
 - VLAN Number: 20
 - VLAN Name: Support
 - Click Add.

4.Verify the VLAN Configuration : Verification Step.

1. In the switch CLI, run the verification command:

```
Switch# show vlan brief
```

2. Look at the output table to verify 10 Sales and 20 Support are listed as active.

Task 2 :- Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

ANS :-

To assign three ports to VLAN 10 (Sales) and verify the configuration, execute the following commands in the switch CLI :-

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range fastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# end
```

Verification Command :-

Run show vlan brief to check the updated VLAN membership table :

```
Switch# show vlan brief
```

Expected Output :-

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Verification Check : Confirm that Fa0/1, Fa0/2, Fa0/3 appear in the Ports column next to 10 Sales and are no longer listed under VLAN 1 (default).

To assign two ports to VLAN 20 (Support) and verify the configuration, run the following commands in the switch CLI :

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range fastEthernet 0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end
```

Verification Command :-

```
Switch# show vlan brief
```

Expected Output :-

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Verification Check : Confirm that Fa0/4, Fa0/5 are displayed in the Ports column next to 20 Support.

Task 3 :- Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router

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ANS :-

Here is how to connect two routers, assign IP addresses to their connecting interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2 in Cisco Packet Tracer.

Network Addressing Plan

- Router 1 ↔ Router 2 Link: 10.0.0.0/30
 - Router 1 (GigabitEthernet 0/0/0): 10.0.0.1
 - Router 2 (GigabitEthernet 0/0/0): 10.0.0.2
- LAN behind Router 2: 192.168.2.0/24
 - Router 2 (GigabitEthernet 0/0/1): 192.168.2.1

1.Connect the Routers:Topology Setup.

1. Drag two 4331 or 2911 Routers onto the workspace (Router0 and Router1).
2. Select Connections (Lightning icon) → Copper Cross-Over cable (dashed line) or Copper Straight-Through.
3. Connect Router 1 (GigabitEthernet 0/0/0) directly to Router 2 (GigabitEthernet 0/0/0).

- *Note:* If using Serial links instead, use a Serial DCE cable between Serial 0/1/0 ports on both routers.

2.Configure Interface IP on Router 1 : Router 1 CLI.

Open Router 1 CLI and configure the link interface:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

3. Configure Interface IPs on Router 2 : Router 2 CLI.

Open **Router 2** CLI and configure both the link interface and its internal LAN interface:

```
Router> enable
Router# configure terminal

! Configure Link to Router 1
Router(config)# interface gigabitEthernet 0/0/0
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

! Configure Local LAN Interface
Router(config)# interface gigabitEthernet 0/0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

4. Configure Static Route on Router 1: Static Routing.

On Router 1, add a static route directing traffic for the 192.168.2.0/24 subnet toward Router 2's IP (10.0.0.2):

```
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)# end
```

5. Verify the Route and Connectivity : Verification.

1. Check Router 1's routing table:

```
Router# show ip route
```

2. Test reachability to Router 2's LAN gateway:

```
Router# ping 192.168.2.1
```

Verification Check :-

When you run show ip route on Router 1, confirm that a entry marked with S (Static) appears:

```
S    192.168.2.0/24 [1/0] via 10.0.0.2
```

A successful ping 192.168.2.1 with a !!!!! result verifies the static route is forwarding traffic correctly.

Task 4 :- Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

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Why the Ping Fails :-

- VLANs create boundaries : VLAN 10 (Sales) and VLAN 20 (Support) act like two separate physical networks. A standard switch blocks direct traffic between them.
- Different Subnets : Devices in Sales (192.168.10.x) and Support (192.168.20.x) speak on different IP ranges.
- No Router : Switches only pass traffic within the *same* VLAN. You need a router to bridge traffic between *different* subnets.

The Fix : Add a Router ("Router-on-a-Stick")

To let Sales and Support talk to each other, connect your switch to a router:

1. Set the switch-to-router port to Trunk mode : This allows traffic from both VLANs to pass through a single cable.
2. Create subinterfaces on the router : Give the router an IP address in VLAN 10 (192.168.10.1) and VLAN 20 (192.168.20.1) to act as default gateways.
3. Point PCs to the router : Set each PC's Default Gateway to its VLAN's router IP.

Now, when Sales pings Support, the traffic goes up to the router, gets routed to VLAN 20, and arrives at Support.